

Amazon Sales Dataset

Data Cleaning & Reporting Automation Project

This report documents an end-to-end data cleaning, exploratory analysis, and reporting automation pipeline built on the Amazon Sales product dataset (1,465 raw rows, 16 columns), prepared as part of a Data Analysis internship task.

Dataset Summary

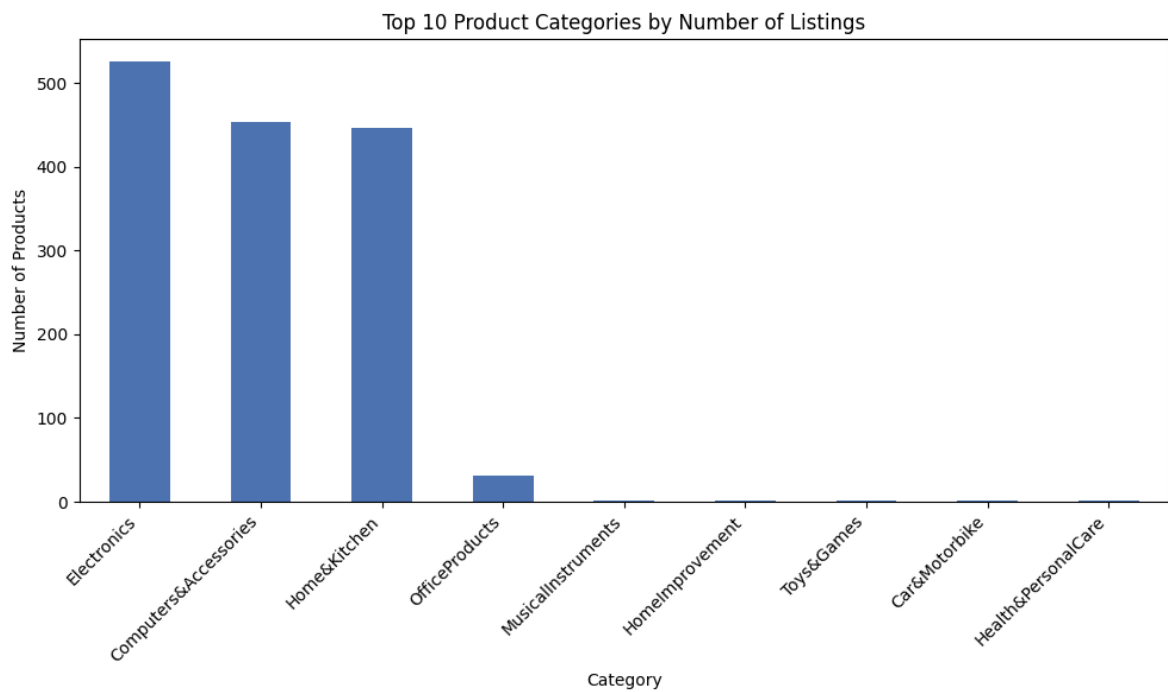
Metric	Value
Raw rows	1465
Duplicate rows removed	0
Cleaned rows	1464
Total categories	9
Average rating	4.1
Average discount %	47.71%
Average discounted price	■3126.01
Average actual price	■5447.0

Key Findings

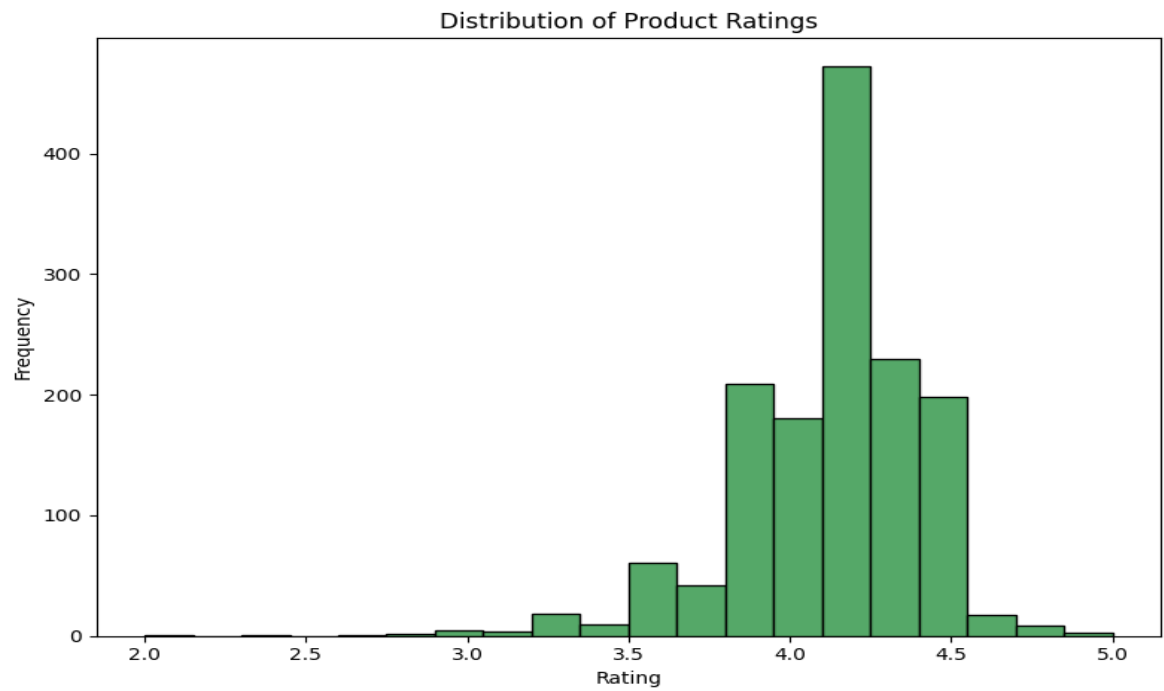
- The largest product category is **Electronics**, with 526 listed products.
- Average customer rating across all products is **4.1** out of 5, with ratings ranging from 2.0 to 5.0.
- Products carry an average discount of **47.71%**, with the steepest discount reaching 94.0%.
- The most-reviewed product is **AmazonBasics Flexible Premium HDMI Cable (Black, 4K@60Hz, 18**.
- The most expensive product listed is **Sony Bravia 164 cm (65 inches) 4K Ultra HD Smart LED Google** at ■139,900.

Charts & Visualizations

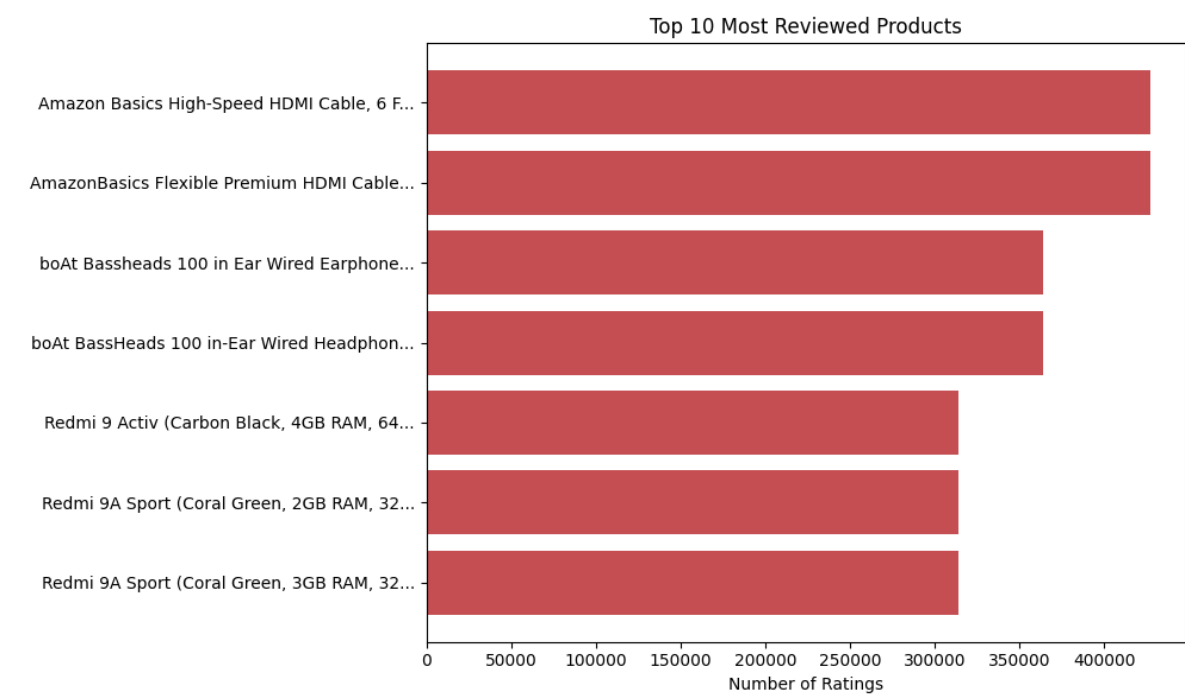
Top 10 Categories by Number of Listings



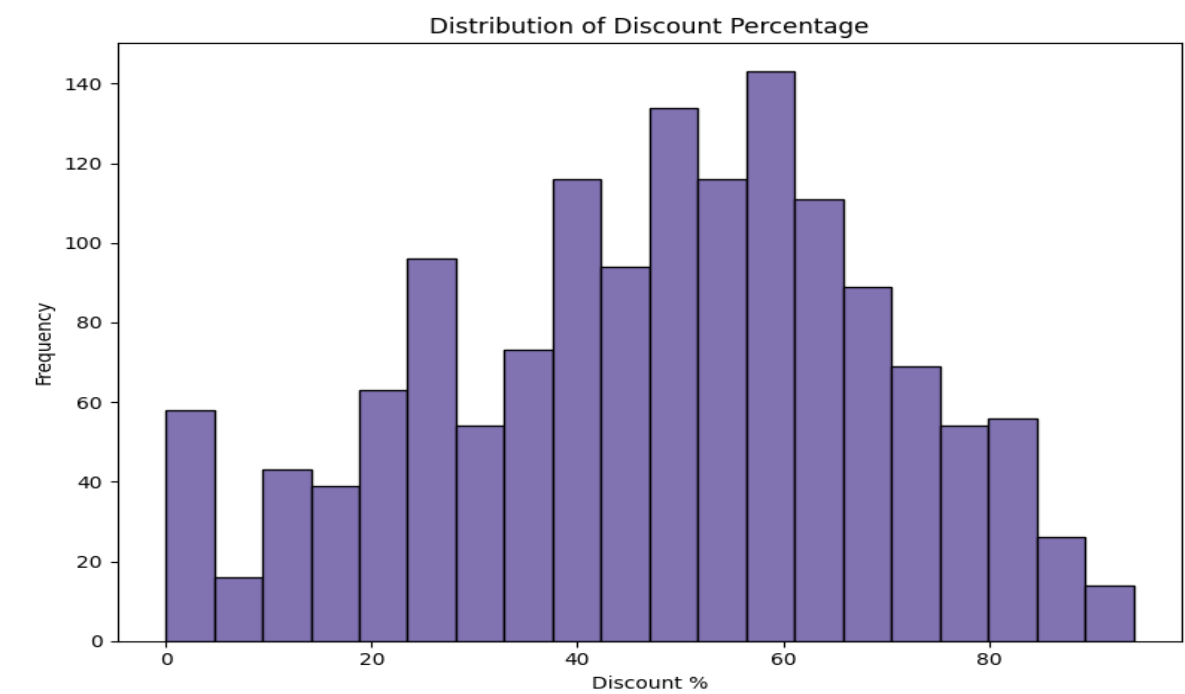
Rating Distribution



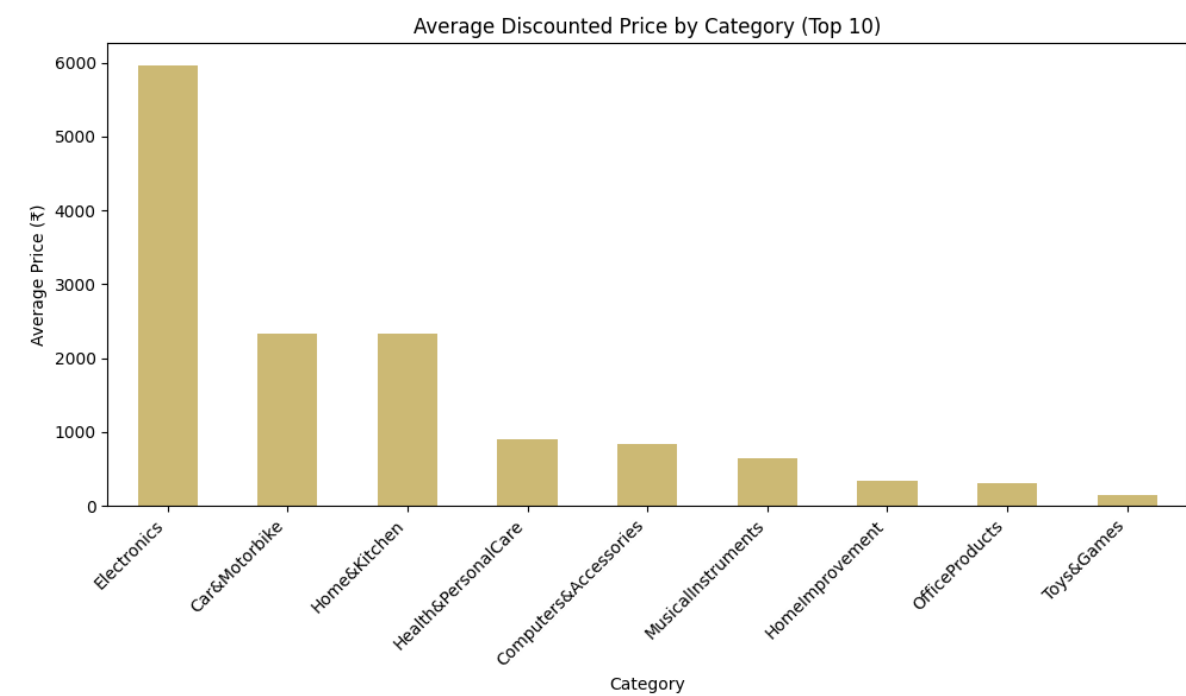
Top 10 Most Reviewed Products



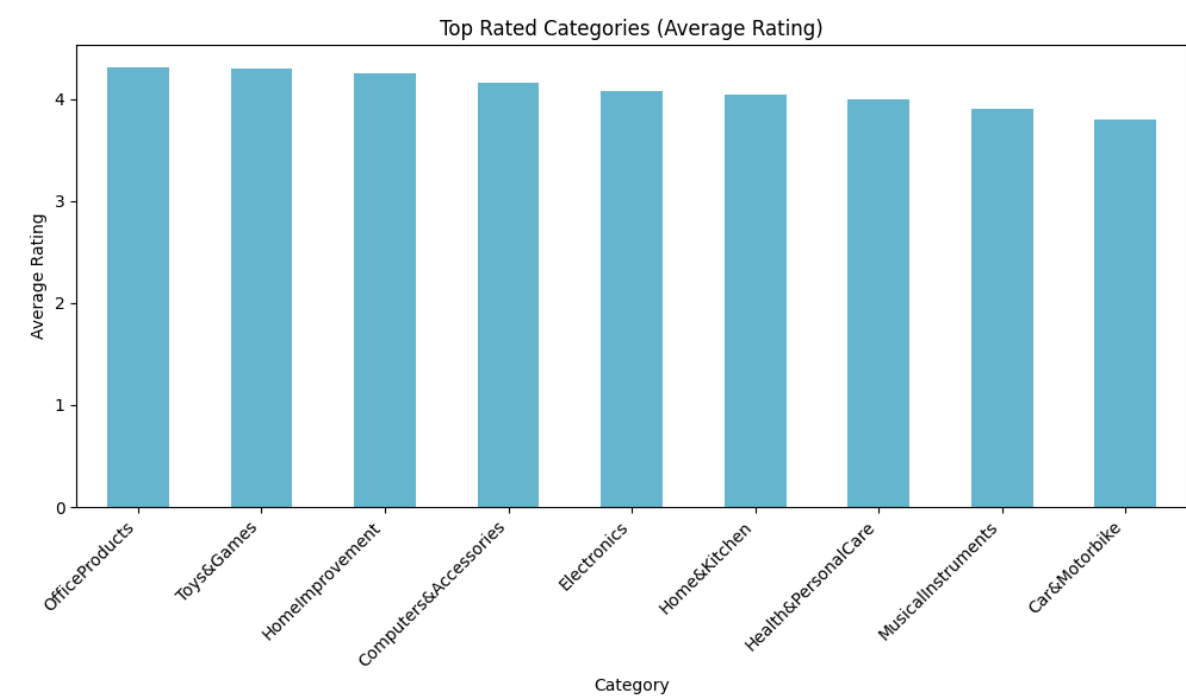
Discount Percentage Distribution



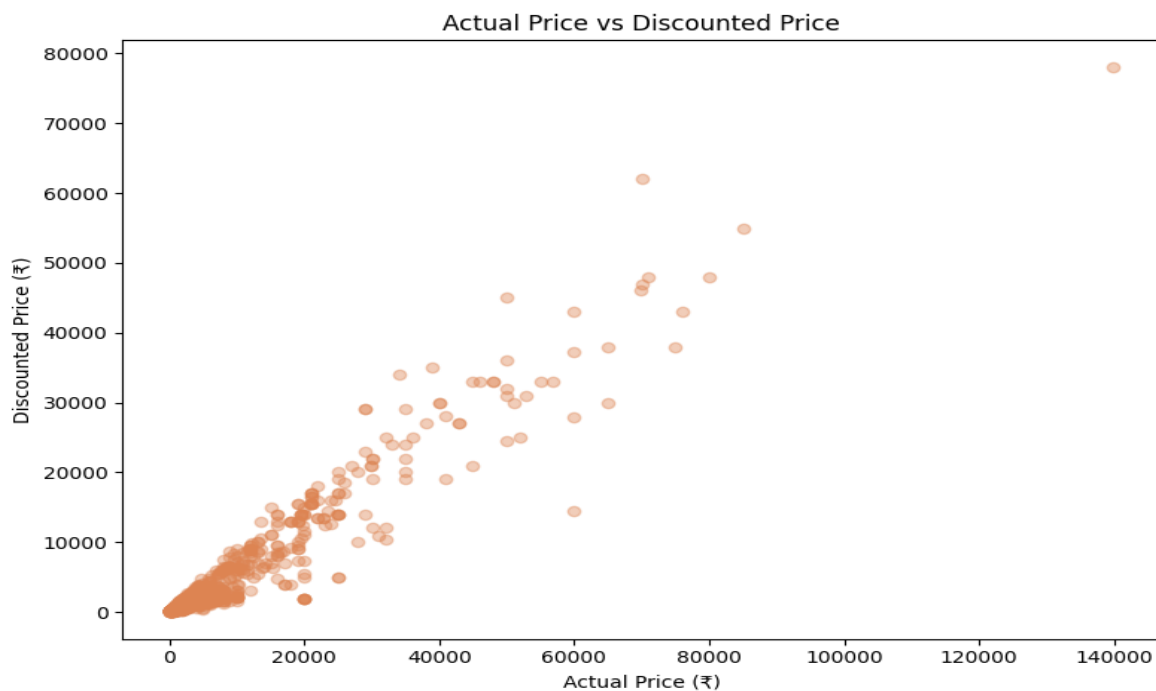
Average Price by Category



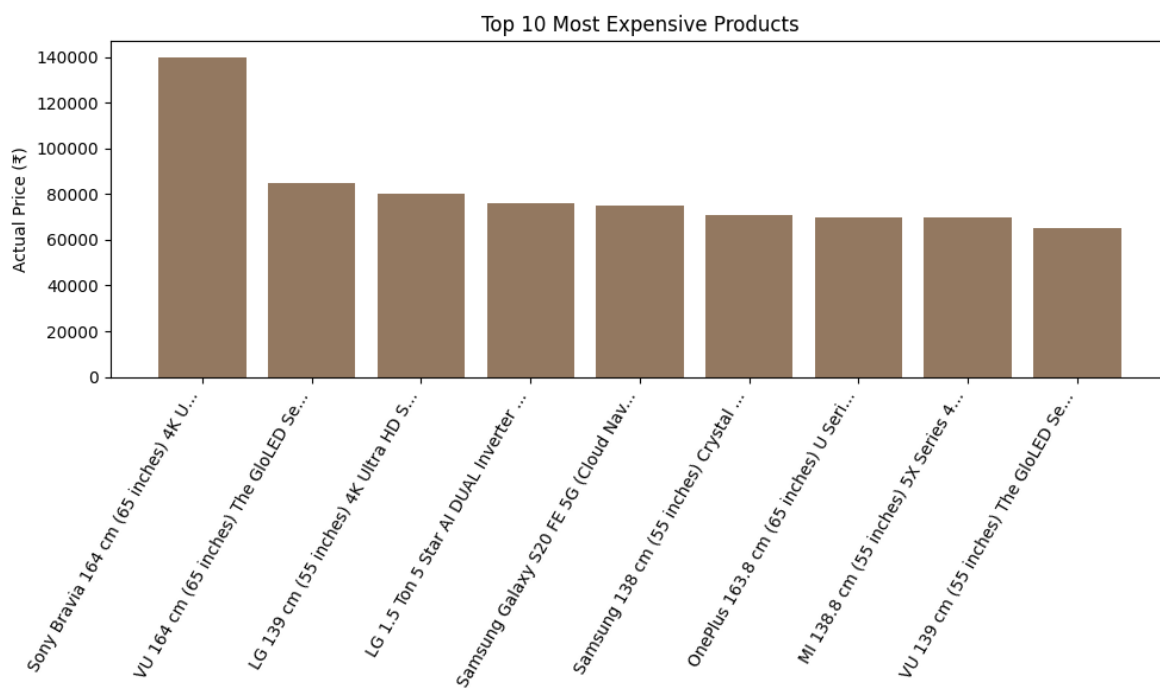
Top Rated Categories



Actual Price vs Discounted Price



Top 10 Most Expensive Products



Conclusion

The raw Amazon sales dataset contained duplicate rows, currency symbols and thousands separators embedded in numeric fields, inconsistent text spacing, and a small number of invalid rating values. After cleaning, the dataset was reduced to a consistent, analysis-ready table. The automated pipeline (`main.py`) can be re-run on any updated export of this dataset to regenerate the cleaned CSV, charts, and this report without manual intervention.